

Minjee Kim

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Sufficient dimension reduction (SDR) for complex data structures, with applications in biomedical statistics

EDUCATION

- Sep 2022 – Feb 2027 **Ewha Womans University**
- *Ph.D.* in Statistics (Advisor: **Jae Keun Yoo**)
T.A.: Regression Analysis, Bayesian Statistics, Mathematical Statistics, etc.
- Mar 2017 – Aug 2022 • *B.S.* in Architectural and Urban Systems Engineering / Statistics
- Jan 2025 – Jun 2025 **University of Toronto**
- *Visiting Student* in Mechanical & Industrial Engineering

RESEARCH EXPERIENCES

- Sep 2022 – Feb 2027 **Ewha Dimension Reduction and Intelligent Learning Lab**
Graduate Student Researcher
- Extended SDR methods to functional data, survival regression, non-elliptical settings
- Jan 2025 – Jun 2025 **Sunnybrook Health Sciences Centre**
Visiting Researcher
- Developed a dynamic mortality prediction model for prostate cancer patients

PUBLICATIONS

1. **Non-elliptical Dimension Reduction in Survival Regression**,
Minjee Kim, Minjeong Kim and Jae Keun Yoo, 2026, *Stat*, 15(3), e70162.
2. **Dynamic Mortality Prediction for Prostate Cancer based on Irregularly Sampled Multimodal EHR and Musculoskeletal Image-based Biomarkers**,
Eunseon Seong, Minjee Kim, Dareen Eom, Gyeonghwa Heo, Urban Emmenegger, Saleh Tabatabaei, Tayler Declan Ross, and Michael Hardisty, 2026, In *Pacific-Asia Conference on Knowledge Discovery and Data Mining* (pp. 399-410).
3. **Nonlinear Functional Sufficient Dimension Reduction via Principal Fitted Components**,
Minjee Kim, Yujin Park, Kyongwon Kim, and Jae Keun Yoo, 2025, *Statistics and Computing*, 35(4), 95.
4. **Covariance Projective Resampling Informative Predictor Subspace for Multivariate Regression**,
Minjee Kim & Jae Keun Yoo, 2025, *Journal of the Korean Statistical Society*, 54(3), 920-931.
5. **Applications of response dimension reduction in large p-small n problems**,
Minjee Kim & Jae Keun Yoo. 2024, *Communications for Statistical Applications and Methods*, 31(2), 191-202.

IN PREPARATION

1. **Second-Order Covariance Projective Resampling for Sufficient Dimension Reduction with Multivariate Responses**, Minjee Kim & Jae Keun Yoo.

CONFERENCES

- Jun 2026 **30th PAKDD** – “Dynamic Mortality Prediction for Prostate Cancer using Multimodal EHR”
- Dec 2025 **13th IASC-ARS** – “CovPRIPS for Multivariate Regression”
- Aug 2025 **8th EcoSta** – “Nonlinear Functional SDR via Principal Fitted Components”
- Dec 2024 **IASC-ARS Interim** – “Response dimension reduction in large p-small n problems”

SKILLS

- **R** (primary), **Python / SQL** (Clinical EHR data querying and preprocessing)
- Languages – English (academic), Korean (native)

HONORS & ACTIVITIES

Jul 2026	<i>Volunteer</i> of 33rd International Biometric Conference
Aug 2025	Ewha Womans University <i>Best Paper Award</i>
Sep 2021 – Aug 2022	<i>Member</i> of Ewha Statistics Analysis Association